

Broc Weselmann

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EDUCATION

- 2024 - May 2027 B.S. in Computer Science at **University of Central Florida**
Focus: Machine Learning, Transformer Models, Mechanistic Interpretability
- 2021 - 2024 Associate's Degree at **Daytona State College**

PROJECTS

- Encyclopedia AI** — Solo Developer July 2025 - Present
- Architected a **multi-agent LLM pipeline in Go** that plans, drafts, critiques, and revises articles autonomously.
 - Designed a provider layer for local **Ollama** and OpenAI-compatible APIs with structured output, reasoning-model streaming, and retries; ran generation as resumable background jobs persisted in **SQLite**.

- SINEBot** — Solo Developer June 2026 - Present
- Built a **JavaScript** Discord bot for UCF's Society of Innovation and Neurodiversity in Engineering (SINE), serving **about 20 daily active users** with a **plugin registry** of daily games and Chart.js analytics.
 - Shipped features iteratively from user feedback, including leaderboards, trend graphs, and distributions.

- MERN Travel Agent** — Backend & DevOps, Team Project March 2026 - April 2026
- Set up the repo, **MongoDB** database, JWT auth, and bcrypt hashing for a **React/TypeScript** web app.
 - Automated **DigitalOcean** deploys with a **GitHub Actions** CI/CD pipeline and PR integration tests.

- Drone Route Optimizer** — Algorithm Developer, KnightHacks VIII October 2025
- Designed the routing algorithm for NextEra Energy's challenge using VRP heuristics and **OR-Tools**.

- GPT-1 Language Model** — Solo Developer June 2026
- Trained a GPT-1 transformer in **PyTorch** from Karpathy's lecture, adding scaled residuals and checkpointing.

- A Frail Qubit** — Solo Developer January 2019 - Present
- Developed a story-based puzzle game end-to-end in the **Godot Engine** with **GDScript**, creating all mechanics, narrative, pixel art, and **over 60 handcrafted levels**.

- The New Order: The Last Days of Europe** — Senior Contributor January 2020 - January 2022
- Implemented event chains, GUIs, and content for a total-overhaul game mod with **over 500,000 players**, and authored the contributor Git guide used by volunteers worldwide.

WORK EXPERIENCE

- Publix** — Grocery Clerk August 2021 - Present
- Promoted from **Front Service Clerk to Cashier to Grocery Clerk**, cross-training at two stores.
 - Unloaded shipments, stocked shelves, and maintained aisles while averaging **30 hours per week** during school.

SKILLS

- Languages:** Python, Go, TypeScript, C, GDScript
- Frameworks & Data:** React, Node.js, Express, PyTorch, NumPy, SQLite, MongoDB, MySQL
- Agentic AI:** Claude Code, Codex, multi-agent LLM pipelines, Ollama
- Tools & Platforms:** Git, GitHub Actions, GNU/Linux, DigitalOcean, Godot Engine